JORGE ALEJANDRO PARRA BENAVIDES

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LinkedIn | GitHub | Portfolio

Professional Summary

Software Engineering student focused on full-stack development, software architecture, and building robust systems. Academic project experience designing and integrating applications with Java, Spring Boot, TypeScript, and PostgreSQL, from data modeling to deployment. Interested in maintainable solutions, security, automated testing, and pragmatic problem-solving.

Education

Universidad Cooperativa de Colombia Pasto, Colombia

Software Engineering (in progress) *2024 – 2028 (expected)*

Technical Skills

Languages: Java, JavaScript, Python, TypeScript, C#, Kotlin, HTML/CSS.

Frameworks: Spring Boot, React, Next.js, Node.js, Angular, Django, .NET, Jetpack Compose.

Data and infrastructure: PostgreSQL, MySQL, MongoDB, Supabase, Docker, AWS, GitHub Actions, Vercel.

Tools and practices: Git, GitHub, Postman, Swagger/OpenAPI, Maven, REST APIs, OOP, microservices, Scrum.

Selected Academic Projects

Ganadería 4.0 | *Java, Spring Boot, PostgreSQL, Docker* Mar. – May 2026

- Designed and developed the REST backend for an IoT livestock monitoring system, managing animals, collars, geofences, GPS telemetry, and operational alerts.
- Implemented JWT authentication with four roles, HMAC SHA-256 signatures with replay protection, and notifications; set up CI, deployment on Render, and **475 documented automated tests**.

Vision2Code | *Tauri, React, TypeScript, NestJS, PostgreSQL* Sep. – Nov. 2026

- Architected a monorepo and developed a Windows application that converts UI screenshots into editable React and Tailwind CSS projects using AI-assisted analysis and deterministic code generation.
- Implemented backend services, authentication, and data persistence; configured CI and preview security controls, with **978 reported automated tests** and a working Windows installer.

ZeroWaste-Eats | *Kotlin, Jetpack Compose, NestJS, Supabase* Oct. – Nov. 2026

- Designed the architecture and led integration of a native Android app to reduce food waste through nearby offers, reservations, pickup codes, and AI-assisted listing suggestions.
- Integrated authentication, row-level security (RLS) policies, and AI usage limits; oversaw testing on a physical Android device and backend deployment, with **over 1,000 reported automated tests**.

Additional Training

Attended four Software Engineering seminars organized by Universidad Cooperativa de Colombia, including the *National Software Engineering Seminar: The Software Development Industry in Colombia*.

Languages

Spanish: native. English: B2 (self-reported).